

AI Customer Support Agent for AmazonHelp

Hiver SDE Intern Take-Home Assignment | Brand: AmazonHelp | Dataset: Customer Support on Twitter

1. Problem Framing, Scope, and Definition of “Good”

The assignment asks for an AI agent that can support AmazonHelp, Amazon's customer-service Twitter handle, by classifying an incoming message's intent, retrieving relevant historical resolutions, drafting a grounded reply, and deciding whether the case is safe to auto handle or must be escalated to a human agent. The central engineering tension is that an incorrectly auto-handled reply (a false auto-handle) can cause real financial, security, or safety harm, while an unnecessary escalation only costs human review time. The system is therefore designed around a safety first escalation policy rather than around maximizing raw accuracy.

A good AmazonHelp agent must: correctly understand the customer's issue; retrieve relevant historical evidence rather than guessing; draft a helpful, grounded response that avoids unsupported claims; avoid exposing historical artifacts (names, agent handles, internal links) to customers; escalate risky cases (financial disputes, security, account access); and default to human review whenever confidence or retrieved evidence is insufficient. Accuracy alone is not the goal safe, evidenced, and appropriately escalated behavior is.

Deliberately not built: live Amazon account integration, payment processing, browser automation, a production ticketing system, a full production UI, and a full conversational-memory service. These were out of scope for a take-home and would not change the evaluation methodology below.

2. Architecture

Pipeline: customer message → intent classification → historical retrieval → escalation decision → grounded reply generation → structured output.

- **Intent classification:** semantic centroid / prototype matching over 8 support intents, with temperature-calibrated confidence rather than raw cosine scores.
- **Retrieval:** dense retrieval using all-MiniLM-L6-v2 embeddings with cosine-similarity ranking over historical AmazonHelp conversations.
- **Escalation:** rule-based, safety-first gating (financial disputes, account/security, low confidence, weak evidence) evaluated before reply generation, so the decision can shape reply wording.
- **Reply generation:** grounded synthesis constrained to verified historical precedents, with anti-hallucination safeguards and sanitization of historical artifacts (customer names, agent tags, internal links) before reuse.
- **Output:** structured record containing intent, confidence, decision, reason, retrieved precedents, and draft reply.

3. Golden Evaluation Set and Leakage Control

The golden set contains 200 hand-labelled AmazonHelp examples, balanced at 25 examples across each of 8 intents (delivery_delay_tracking, refund_return_request, damaged_defective_item, payment_billing_issue, account_login_access, subscription_prime_issue, product_inquiry_availability, general_complaint_feedback). Examples were sampled from the Customer Support on Twitter AmazonHelp subset to cover routine self-service requests alongside higher-risk cases such as account security and financial disputes, then labelled with a gold intent, a gold auto-handle/escalate decision, and an expected resolution procedure. **Leakage control:** all 200 golden messages and their conversation IDs were removed from the historical retrieval corpus, and the evaluator passes golden conversation IDs into the retriever's exclusion filter; verified overlap between the golden set and the retrieval corpus is 0.

4. Evaluation Harness: Automated Metrics and LLM-as-Judge

Automated metrics (intent accuracy, macro F1, escalation accuracy/recall, dangerous false auto-handles, retrieval Recall@3) are computed directly against the golden labels. Reply quality is additionally scored by an LLM-as-judge rubric, overall rated 4.42 / 5.0. To check whether the judge is trustworthy, 40 cases were independently scored by a human and compared to the judge: Spearman $\rho = 0.3844$, Pearson $r = 0.3134$, mean absolute difference (MAD) = 0.167, and 95.0% near-agreement (judge and human within a small tolerance band). These correlations indicate **moderate, limited agreement** — not strong or high agreement — so the LLM judge is used as a supporting signal for reply-quality trends, not as ground truth for pass/fail decisions.

5. Results vs. Baselines

System	Intent Acc.	Macro F1	Escalation Acc.	Escalation Recall	Dangerous False Auto-Handles
Majority (trivial baseline)	12.50%	0.0278	66.50%	0.00%	67 / 67 (100%)
TF-IDF + Logistic Regression (simple baseline)	44.50%	0.4344	65.50%	26.87%	49 / 67 (73.1%)
AI Agent	77.00%	0.7650	61.50%	91.04%	06 / 67 (8.96%)

The agent improves intent accuracy by 32.5 points over the TF-IDF baseline and by 64.5 points over the majority baseline, reaching 0.7650 macro F1. Escalation recall of 91.04% cuts dangerous false auto-handles to 6 of 67 (vs. 49/67 and 67/67 for the baselines). The agent's overall escalation accuracy (61.50%) is lower than the majority baseline's (66.50%) by design: the majority baseline achieves accuracy by escalating almost everything, which is safe but has no automation value, whereas the agent trades some accuracy for a large gain in recall on the cases that actually matter. Supporting metrics: retrieval Recall@3 = 45.5%; LLM judge overall rating = 4.42/5.0 (see §4 for judge-agreement caveats).

6. What Is Misleading About My Headline Number?

- 77% intent accuracy is **not** 77% end-to-end customer resolution quality — it only measures whether the predicted intent label matched the gold label.
- The golden benchmark has only 200 examples, and the held-out test partition used for headline numbers has just 60 examples, so a handful of cases can swing the percentage noticeably.
- The benchmark is intentionally **balanced** at 25 examples per intent; production traffic is not balanced and skews heavily toward delivery-tracking questions, so this number may not transfer directly.
- Intent accuracy says nothing about factual correctness, actionability, empathy, or whether the customer's underlying problem was actually resolved.
- Retrieval can return precedents that are semantically related but operationally different from the current case, which can mislead reply generation even when the intent is correct.
- The historical Twitter support corpus contains generic or canned responses, so retrieved evidence can bias replies toward deflection rather than resolution.
- Escalation risk is **asymmetric**: a false auto-handle can cause real financial, security, or safety harm, while a false escalation mainly adds human workload — a single "decision accuracy" number hides this asymmetry.
- LLM-judge scores (4.42/5.0) are not equivalent to human ground truth; agreement with human ratings is moderate and limited (§4), so the judge score can reward polished tone even when a human would flag the answer as too generic.
- The 8-intent taxonomy cannot represent every real production case, so fallback escalation for out-of-taxonomy messages remains necessary.
- The dangerous-false-auto-handle rate (6/67) is operationally meaningful but is not a universal north star either: escalating every case would drive it to zero while eliminating all automation value.

7. Top 5 Failure Modes (Real Evaluation Examples)

Failure modes below were identified directly from actual evaluation predictions, not chosen as convenient categories in advance.

7.1 Lexical Saliency Bias

Customer message (Case #14): “Does Amazon Prime deliver packages on Sunday in postal code 98101?”

Predicted: subscription_prime_issue (conf. 0.88)	Gold: delivery_delay_tracking	Decision: AUTO_HANDLE
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Observed behavior: The token “Prime” became overly salient to the classifier and overshadowed the actual question, which was about Sunday delivery, not subscription management.

Hypothesis / next step: Add contrastive training examples that separate “Prime” used as a delivery modifier from “Prime” used to describe subscription actions, and give the classifier better semantic/token-role handling.

7.2 Entity Mention Misattribution

Customer message (Case #19): “How do I add a gate code so the driver can access my apartment building?”

Predicted: general_complaint_feedback (conf. 0.88)	Gold: delivery_delay_tracking	Decision: ESCALATE
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Observed behavior: The word “driver” was associated with historical driver-misconduct complaints in the embedding space, even though this message is a neutral delivery-configuration question.

Hypothesis / next step: Use question-form and sentiment/context signals plus contrastive delivery-vs.-misconduct examples so entity mentions are not treated as intent on their own.

7.3 Compound and State-Transition Query

Customer message (Case #15): “My tracking status says ‘Returned to sender - Damaged in transit’. What happens now?”

Predicted: damaged_defective_item (conf. 0.88)	Gold: delivery_delay_tracking	Decision: AUTO_HANDLE
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Observed behavior: “Damaged” was interpreted as a physical product defect, even though the customer never received the item — the damage occurred pre-delivery, in transit.

Hypothesis / next step: Model fulfillment state explicitly and distinguish pre-delivery carrier damage from post-delivery product defects.

7.4 Subtle Verbal Grievance Bypassing Escalation

Customer message (Case #189): “The customer service representative could barely understand basic English and hung up.”

Predicted: general_complaint_feedback (conf. 0.63); top historical similarity 0.535	Gold: ESCALATE (gold decision)	Decision: System decision: AUTO_HANDLE — dangerous false auto-handle
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Observed behavior: A subtle service-quality grievance (agent hung up on the customer) was scored as a routine complaint and auto-handled instead of routed for supervisory review.

Hypothesis / next step: Expand conduct-escalation triggers to include phrases such as “hung up,” “disconnected,” and other communication-barrier signals as supervisory review triggers.

7.5 Borderline Uncertainty Safety Escalation

Customer message (Case #2): “My order has been stuck in ‘Departed Facility’ in Memphis for 5 days with no new scan.”

Predicted: delivery_delay_tracking (conf. 0.48); top historical similarity 0.573	Gold: AUTO_HANDLE (gold decision)	Decision: System decision: ESCALATE (unnecessary escalation)
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Observed behavior: The predicted intent was correct and retrieval was well-grounded, but confidence (0.48) fell just under the fixed 0.50 auto-handle threshold, triggering an unneeded escalation.

Hypothesis / next step: Replace the absolute confidence threshold with a decision that also uses the confidence margin over the runner-up intent.

8. What I Would Do With One More Week

- Improve difficult intent boundaries identified above (Prime-as-modifier, entity misattribution, state-transition damage) with targeted contrastive examples.
- Tune the auto-handle threshold using confidence margin over the runner-up intent instead of a single absolute cutoff.
- Improve retrieval precision, e.g. secondary reranking by intent consistency between query and retrieved precedent.
- Strengthen escalation triggers for subtle conduct/communication-barrier grievances (failure mode 4).
- Expand human evaluation coverage beyond the current 40-case judge-agreement sample to tighten confidence in the LLM-judge correlation estimates.
- Explicitly measure the automation-utility vs. safety tradeoff (dangerous-false-auto-handle rate vs. % of volume automated) to choose a production operating point rather than a single fixed threshold.

9. Decision Log (Non-Obvious Engineering Decisions)

- Scoped the project to a single brand (AmazonHelp) rather than the full multi-brand Twitter support dataset, to keep the golden set and taxonomy coherent within the assignment timeframe.
- Defined 8 intents empirically from AmazonHelp's actual historical support patterns rather than adopting a generic, off-the-shelf intent taxonomy.
- Balanced the golden benchmark at 25 examples per intent instead of sampling in proportion to natural traffic, to prevent common intents from dominating the accuracy metric.
- Removed golden examples and their conversation IDs from the retrieval corpus and enforced this via an explicit exclusion filter, rather than trusting a train/test split alone, to guarantee zero leakage.
- Used semantic centroid/prototype matching for intent classification instead of training a heavier supervised model, trading a small amount of accuracy for interpretability and fast iteration.
- Added temperature calibration on top of raw cosine similarity so confidence scores are usable as a decision input, not just a ranking signal.
- Chose all-MiniLM-L6-v2 for retrieval embeddings as a lightweight, CPU-friendly model suited to a reproducible take-home rather than a larger embedding model.
- Kept retrieved evidence visible in the structured output (rather than hidden inside the reply) so grounding can be audited case by case.
- Used intent-consistency between the query and retrieved precedents only as a secondary signal, not a hard filter, to avoid discarding useful evidence when the top-level intent is ambiguous.
- Moved the escalation decision *before* reply generation so the safety decision can influence reply wording (e.g., a cautious tone for cases flagged for review).
- Added dedicated escalation and reply-handling logic for financial disputes specifically, rather than treating them like any other payment_billing_issue case.
- Sanitized customer names, agent tags, and historical link artifacts from retrieved precedents before reuse, to avoid leaking irrelevant personal or internal information into new replies.
- Accepted lower overall escalation *accuracy* in exchange for much higher escalation *recall*, a deliberate safety-first tradeoff rather than optimizing a single blended metric.
- Reported LLM-judge/human agreement transparently as moderate and limited (§4), rather than citing the 4.42/5.0 judge score alone, to avoid overstating what the automated reply-quality metric proves.

10. Reproducibility

The repository ships a runnable CLI pipeline, evaluation harness, unit tests, processed historical data, the golden set, and machine-readable evaluation artifacts (Python 3.10+; tested on 3.11). The README documents installation, tests, benchmark execution, and CLI demos, and is designed so the headline results reproduce within the assignment's under-15-minute requirement. Benchmark evaluation itself completes in roughly 16–18 seconds on CPU once model weights are cached.

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python -m pytest tests/ -v
python evaluate.py
python main.py --message "Where is my package?"
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GitHub Repository: <https://github.com/karthikesh21/Hiver-SDE-Intern->